3D WEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

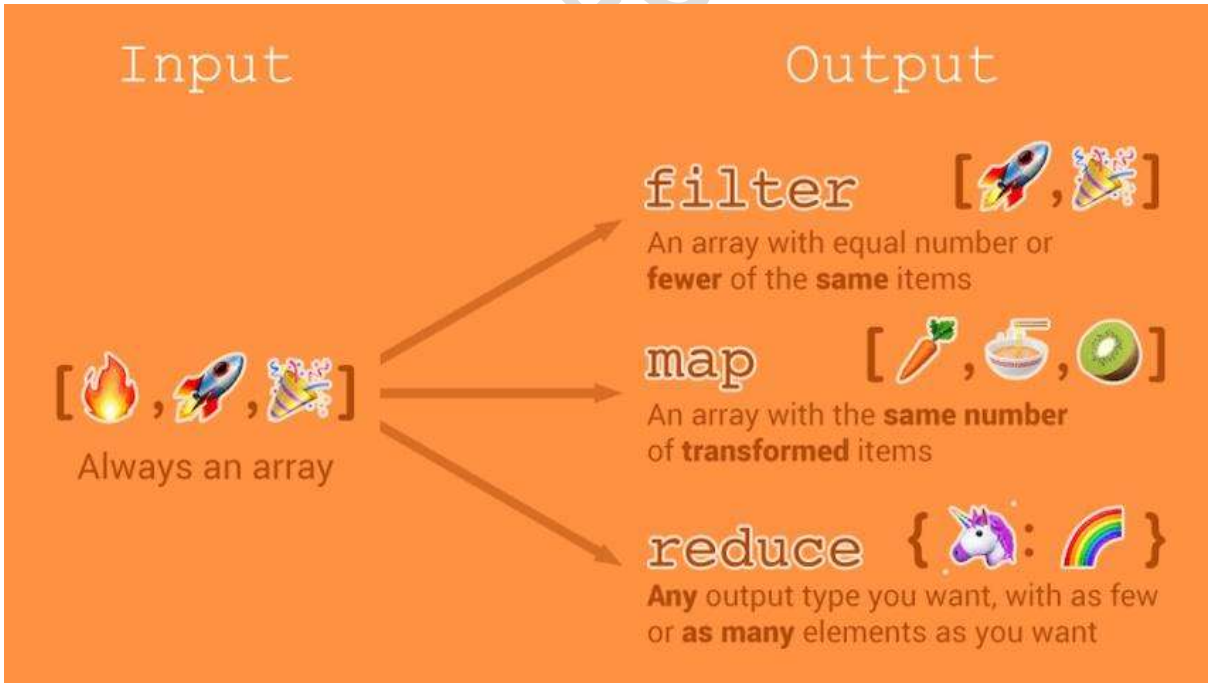
Classes : Offline & Online

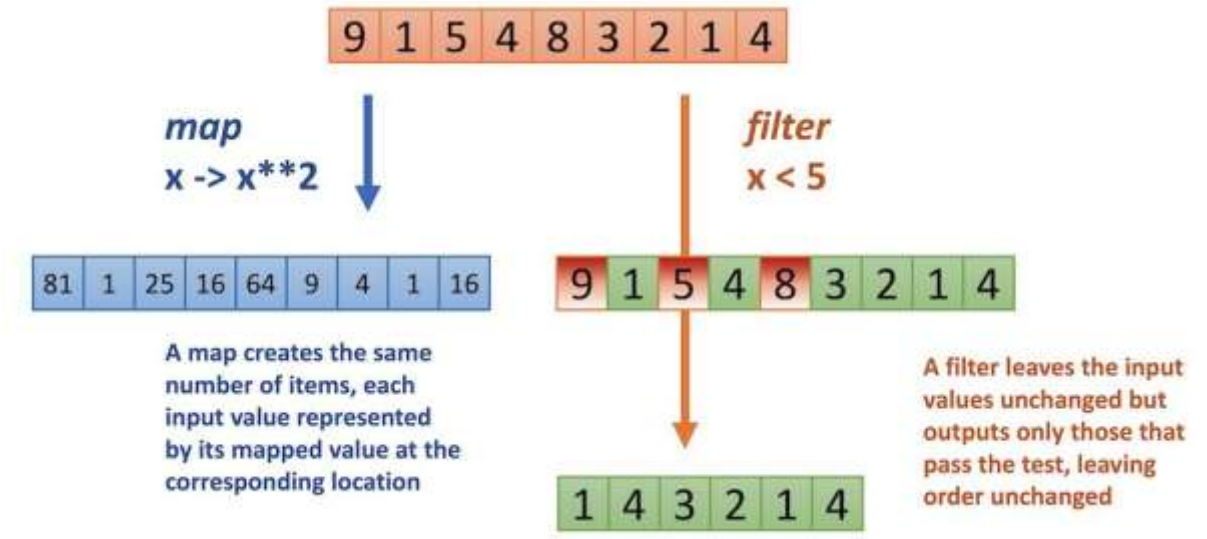
Daily Teaching : 2 Hours

Total Classes : 100

Level 4 – AI Professional Foundation(11th–12th Standard)

Class 12 – map(), filter() & reduce()







1. Learning Objectives

By the end of this lesson, students will be able to:

- Understand map(), filter() and reduce().
- Apply functions to lists and datasets.
- Transform data using map().
- Select required data using filter().

- Calculate a combined result using reduce().
- Use lambda functions with these methods.
- Process simple student, marks and sales datasets.

2. What is Dataset Processing?

A **dataset** is a collection of related data.

For example:

marks = [45, 67, 89, 32, 76, 91]

We may want to:

- increase every mark by 5
- find students who scored above 60
- calculate the total marks

Instead of writing long loops, Python provides useful functions:

DATA

↓

map() → Transform data

↓

filter() → Select required data

↓

reduce() → Combine data into one result

Easy memory trick

3DWEBSOFT FOUNDATION

MAP = Change
FILTER = Select
REDUCE = Combine

3. map() Function

Definition

The map() function applies a function to **every item** in an iterable such as a list.

Syntax

map(function, iterable)

It returns a **map object**, which can commonly be converted into a list.

list(map(function, iterable))

Example 1 – Square Numbers

```
numbers = [1, 2, 3, 4, 5]


squares = list(map(lambda x: x * x, numbers))


print(squares)
```

Output

[1, 4, 9, 16, 25]

How it works

1 → 1 × 1 → 1

3DWEBSOFT FOUNDATION

2 → 2 × 2 → 4

3 → 3 × 3 → 9

4 → 4 × 4 → 16

5 → 5 × 5 → 25

4. map() with Normal Function

```
def double(x):
    return x * 2

numbers = [10, 20, 30, 40]

result = list(map(double, numbers))

print(result)
```

Output

[20, 40, 60, 80]

5. map() Practical Example – Student Marks

Suppose all students receive **5 bonus marks**.

```
marks = [55, 62, 71, 48, 80]

updated_marks = list(map(lambda x: x + 5, marks))
```

3DWEBSOFT FOUNDATION

print(updated_marks)

Output

[60, 67, 76, 53, 85]

Dataset Processing

Original Marks

55 62 71 48 80

↓ ↓ ↓ ↓ ↓

+5 +5 +5 +5 +5

↓ ↓ ↓ ↓ ↓

60 67 76 53 85

6. filter() Function

Definition

filter() is used to **select elements** from a dataset that satisfy a condition.

Syntax

filter(function, iterable)

The function should return:

True → Keep the item

False → Remove the item

Example 1 – Even Numbers

numbers = [1, 2, 3, 4, 5, 6]

3DWEBSOFT FOUNDATION

```
even = list(filter(lambda x: x % 2 == 0, numbers))


print(even)
```

Output

[2, 4, 6]

Working

1 → Odd → Remove

2 → Even → Keep

3 → Odd → Remove

4 → Even → Keep

5 → Odd → Remove

6 → Even → Keep

7. filter() – Student Marks

Find students who scored **60 or above**.

```
marks = [45, 67, 89, 32, 76, 91]


passed = list(filter(lambda x: x >= 60, marks))


print(passed)
```

Output

[67, 89, 76, 91]

3DWEBSOFT FOUNDATION

8. filter() with a Normal Function

```
def is_adult(age):
    return age >= 18

ages = [15, 17, 18, 20, 16, 21]

adults = list(filter(is_adult, ages))

print(adults)
```

Output

[18, 20, 21]

9. reduce() Function

reduce() repeatedly applies a function to the elements of a dataset and produces **one final value**.

It is available in the functools module.

Syntax

```
from functools import reduce

reduce(function, iterable)
```

Example – Find Total

```
from functools import reduce
```

3DWEBSOFT FOUNDATION

```
numbers = [10, 20, 30, 40]

total = reduce(lambda x, y: x + y, numbers)

print(total)
```

Output

100

Working

10 + 20 = 30

30 + 30 = 60

60 + 40 = 100

So:

[10, 20, 30, 40]

↓

reduce()

↓

100

10. reduce() – Find Maximum

```
from functools import reduce

numbers = [25, 72, 41, 90, 63]
```

3DWEBSOFT FOUNDATION

```
maximum = reduce(lambda x, y: x if x > y else y, numbers)


print(maximum)
```

Output

90

11. Difference Between map(), filter() and reduce()

Function	Purpose	Result
map()	Transform every item	Multiple transformed items
filter()	Select items	Selected items
reduce()	Combine items	Usually one value

Remember:

map() → CHANGE

filter() → CHOOSE

reduce() → COMBINE

12. Combining All Three

This is particularly useful for **dataset processing**.

Suppose:

marks = [45, 67, 89, 32, 76, 91]

Step 1 – Add 5 bonus marks using map()

3DWEBSOFT FOUNDATION

```
updated = list(map(lambda x: x + 5, marks))
```

Result:

```
[50, 72, 94, 37, 81, 96]
```

Step 2 – Select marks ≥ 60 using filter()

```
passed = list(filter(lambda x: x >= 60, updated))
```

Result:

```
[72, 94, 81, 96]
```

Step 3 – Calculate total using reduce()

```
from functools import reduce


total = reduce(lambda x, y: x + y, passed)


print(total)
```

Output

343

13. Real-Life Dataset Example – Student Database

```
students = [
    {"name": "Asha", "marks": 85},
    {"name": "Rahul", "marks": 55},
    {"name": "Priya", "marks": 92},
    {"name": "Arjun", "marks": 48}
]
```

SUWEBSOFT FOUNDATION

Find the marks

```
marks = list(map(lambda student: student["marks"], students))


print(marks)
```

Output:

```
[85, 55, 92, 48]
```

Find students who passed

```
passed = list(
    filter(lambda student: student["marks"] >= 60, students)
)


print(passed)
```

Find total marks

```
from functools import reduce


total = reduce(
    lambda x, y: x + y,
    map(lambda student: student["marks"], students)
)


print(total)
```

Output:

```
280
```

3DWEBSOFT FOUNDATION

14. Practical Program – Sales Dataset

```
from functools import reduce

sales = [1200, 2500, 1800, 3200, 1500]

# Increase each sale by 10%
updated_sales = list(map(lambda x: x * 1.10, sales))

# Select sales greater than 2000
large_sales = list(filter(lambda x: x > 2000, updated_sales))

# Calculate total
total = reduce(lambda x, y: x + y, large_sales)

print("Updated Sales:", updated_sales)
print("Large Sales:", large_sales)
print("Total:", total)
```

Expected Output

```
Updated Sales: [1320.0, 2750.0, 1980.0, 3520.0, 1650.0]
Large Sales: [2750.0, 3520.0]
Total: 6270.0
```

15. Important Point – lambda

SUWEBSOFT FOUNDATION

map(), filter() and reduce() are often used with **lambda functions**.

A lambda is a small anonymous function.

Normal function

```
def square(x):
    return x * x
```

Lambda

```
lambda x: x * x
```

Both perform the same operation.

16. Practical Activity

Activity 1 – Square Numbers

Create:

```
numbers = [2, 4, 6, 8, 10]
```

Use map() to produce:

[4, 16, 36, 64, 100]

Activity 2 – Filter Marks

Given:

```
marks = [35, 67, 45, 89, 29, 76]
```

Use filter() to find marks greater than or equal to 50.

Expected:

[67, 89, 76]

3DWEBSOFT FOUNDATION

Activity 3 – Calculate Total

Given:

prices = [100, 250, 150, 300]

Use reduce() to calculate the total.

Expected:

800

17. Mini Project – Student Marks Analyzer

Objective

Create a program that:

1. Stores student marks.
2. Adds 5 bonus marks.
3. Finds students scoring 60 or more.
4. Calculates the total of qualifying marks.
5. Displays the results.

Solution

```
from functools import reduce


marks = [45, 67, 72, 39, 88, 91]


# Add bonus marks
updated_marks = list(map(lambda x: x + 5, marks))
```

3DWEBSOFT FOUNDATION

```
# Select marks >= 60
qualified = list(filter(lambda x: x >= 60, updated_marks))

# Calculate total
total = reduce(lambda x, y: x + y, qualified)

print("Original Marks:", marks)
print("Updated Marks:", updated_marks)
print("Qualified Marks:", qualified)
print("Total:", total)
```

18. Common Mistakes

Mistake 1 – Forgetting list()

```
result = map(lambda x: x * 2, numbers)
```

This gives a map object.

For easy display:

```
result = list(map(lambda x: x * 2, numbers))
```

Mistake 2 – Forgetting the import for reduce()

Incorrect:

```
reduce(lambda x, y: x + y, numbers)
```

Correct:

```
from functools import reduce
```

2DWEBSOFT FOUNDATION

Mistake 3 – Using the wrong condition

filter(lambda x: x > 60, marks)

means **strictly greater than 60**.

Whereas:

filter(lambda x: x >= 60, marks)

means **60 or above**.

19. Quick Revision

map()

Question: What does it do?

Answer: Applies a function to every element.

list(map(lambda x: x * 2, numbers))

filter()

Question: What does it do?

Answer: Selects elements based on a condition.

list(filter(lambda x: x > 50, marks))

reduce()

Question: What does it do?

Answer: Combines multiple elements into one result.

reduce(lambda x, y: x + y, numbers)

20. Knowledge Check

3DWEBSOFT FOUNDATION

1. What is a dataset?
2. What is the purpose of map()?
3. What is the purpose of filter()?
4. Why do we import reduce() from functools?
5. What does lambda mean?
6. What will this produce?

```
list(map(lambda x: x + 10, [1, 2, 3]))
```

7. What will this produce?

```
list(filter(lambda x: x % 2 == 0, [1, 2, 3, 4]))
```

8. Find the output:

```
from functools import reduce


print(reduce(lambda x, y: x + y, [5, 10, 15]))
```

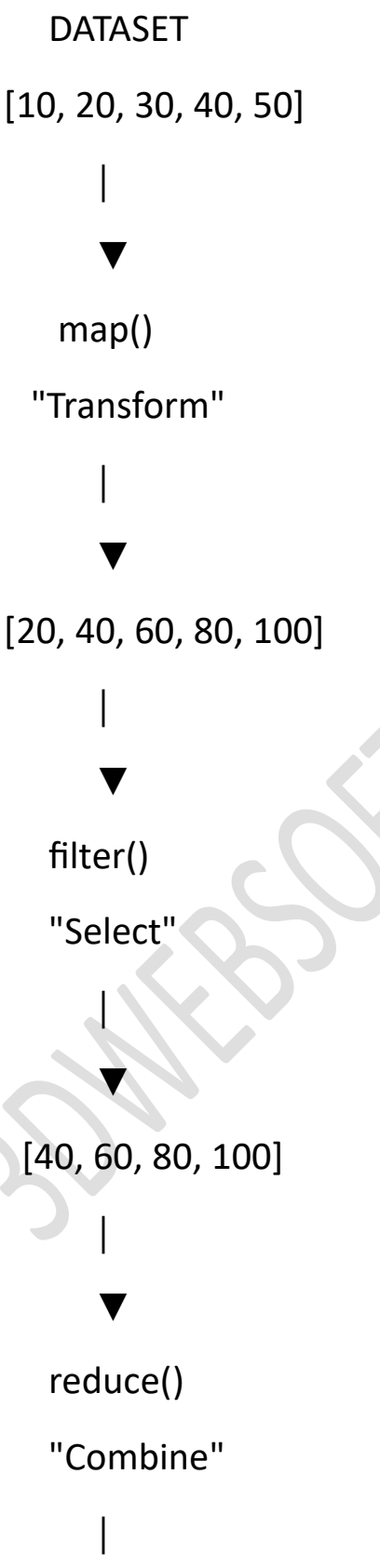
Answers

1. A collection of related data.
2. To transform/process every item.
3. To select items satisfying a condition.
4. reduce() belongs to the functools module.
5. A small anonymous function.
6. [11, 12, 13]
7. [2, 4]
8. 30

3DWEBSOFT FOUNDATION

Teacher Demonstration

Draw this on the classroom board:



3DWEBSOFT FOUNDATION

▼

280

One-line student memory formula:

map() → Modify → filter() → Select → reduce() → Summarize

This lesson connects directly with **Python data processing, datasets, data science and machine-learning preprocessing**, making it a useful foundation for the next lessons.

3DWEBSOFT FOUNDATION